

03

Inverse Trigonometric Functions

1. INTRODUCTION:

The inverse trigonometric functions, denoted by $\sin^{-1}x$ or $(\arcsin x)$, $\cos^{-1}x$ etc., denote the angles whose sine, cosine etc, is equal to x . The angles are usually the numerically smallest angles, except in the case of $\cot^{-1}x$ and if positive & negative angles have same numerical value, the positive angle has been chosen.

It is worthwhile noting that the functions $\sin x$, $\cos x$ etc. are in general not invertible. Their inverse is defined by choosing an appropriate domain & co-domain so that they become invertible. For this reason, the chosen value is usually the simplest and easy to remember.

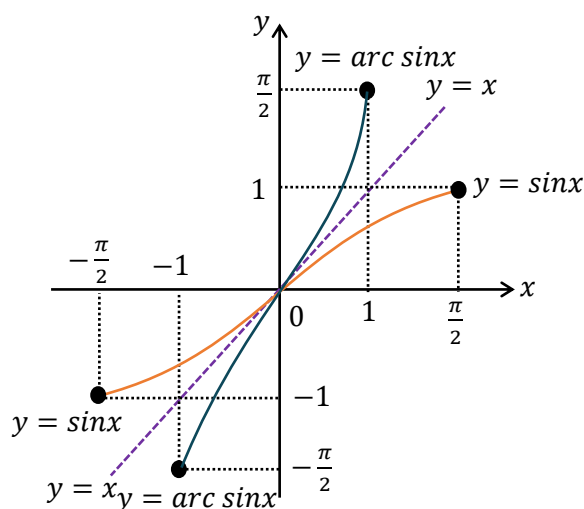
2. GRAPH OF INVERSE TRIGONOMETRIC FUNCTIONS:

$$(a) f : \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \rightarrow [-1, 1]$$

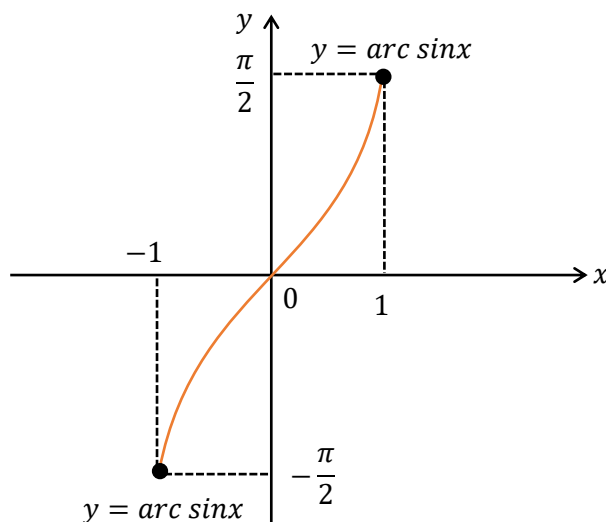
$$f(x) = \sin x$$

$$f^{-1} : [-1, 1] \rightarrow [-\pi/2, \pi/2]$$

$$f^{-1}(x) = \sin^{-1}(x)$$



(Taking image of $\sin x$ about $y = x$ to get $\sin^{-1}x$)



($y = \sin^{-1}x$)

Illustration 1

The value of $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$ is equal to

(A) $\frac{\pi}{4}$

(B) $\frac{5\pi}{12}$

(C) $\frac{3\pi}{4}$

(D) $\frac{13\pi}{12}$

Ans.(C)

Solution:

$$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6} = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$$

Illustration 2

The value of $\sin^{-1}(-\sqrt{3}/2)$ is -

- (A) $-\pi/3$ (B) $-2\pi/3$ (C) $4\pi/3$ (D) $5\pi/3$

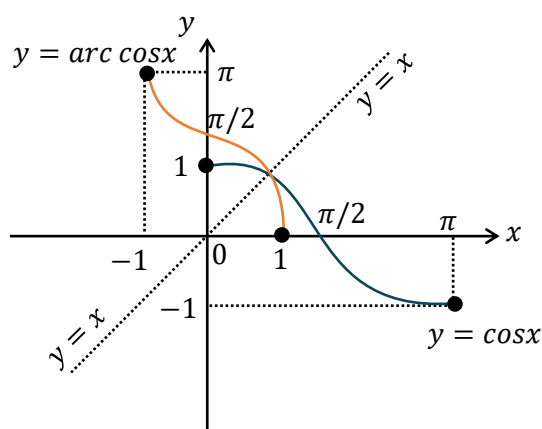
Ans. (A)

Solution:

$$\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) = -\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = -\frac{\pi}{3}$$

(b) $f: [0, \pi] \rightarrow [-1, 1]$

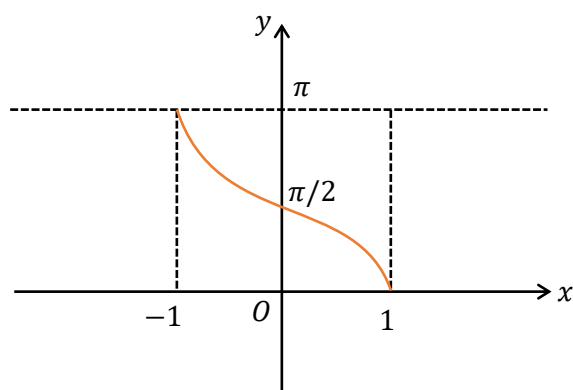
$$f(x) = \cos x$$



(Taking image of $\cos x$ about $y = x$) ($y = \cos^{-1}x$)

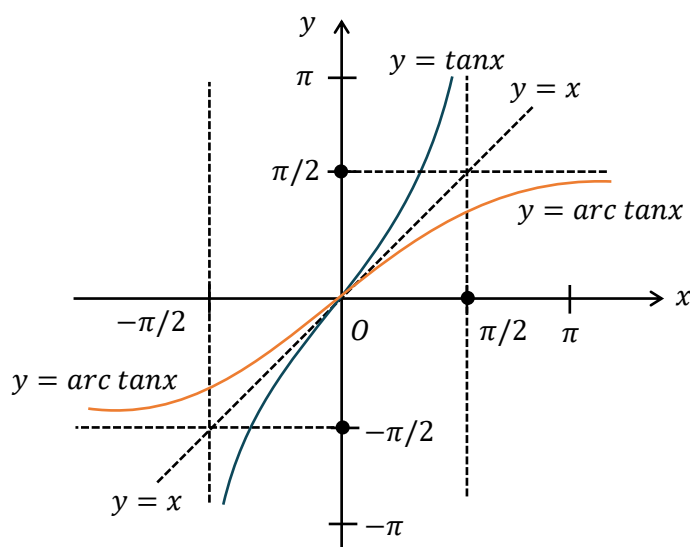
$$f^{-1}: [-1, 1] \rightarrow [0, \pi]$$

$$f^{-1}(x) = \cos^{-1}x$$



(c) $f: (-\pi/2, \pi/2) \rightarrow \mathbb{R}$

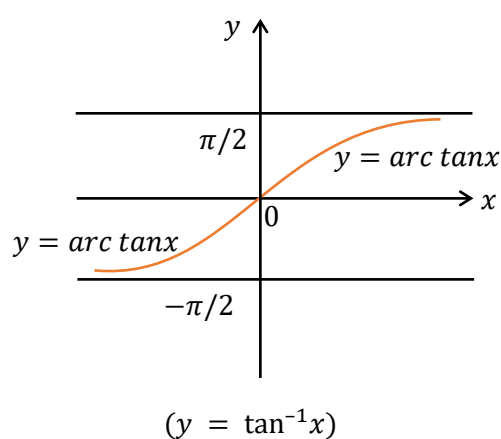
$$f(x) = \tan x$$



(Taking image of $\tan x$ about $y = x$)

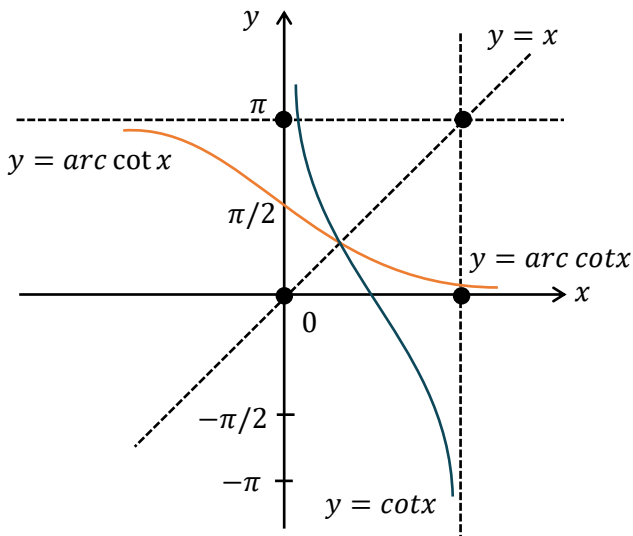
$$f^{-1}: \mathbb{R} \rightarrow (-\pi/2, \pi/2)$$

$$f^{-1}(x) = \tan^{-1}x$$



(d) $f: (0, \pi) \rightarrow \mathbb{R}$

$$f(x) = \cot x$$



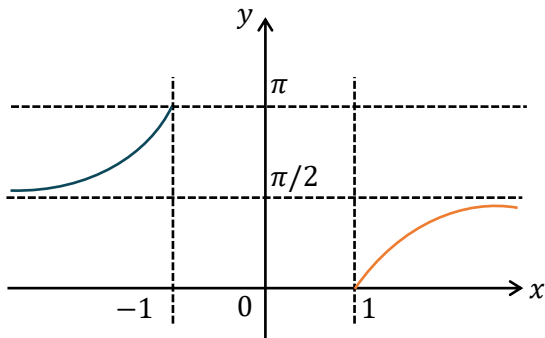
(Taking image of $\cot x$ about $y = x$)

(e) $f: \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right] \rightarrow (-\infty, -1] \cup [1, \infty)$

$$f(x) = \sec x$$

$$f^{-1}: (-\infty, -1] \cup [1, \infty) \rightarrow \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

$$f^{-1}(x) = \sec^{-1} x$$

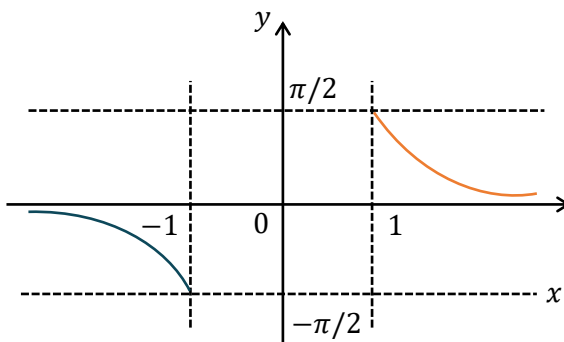


(f) $f: \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right] \rightarrow (-\infty, -1] \cup [1, \infty)$

$$f(x) = \operatorname{cosec} x$$

$$f^{-1}: (-\infty, -1] \cup [1, \infty) \rightarrow \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$$

$$f^{-1}(x) = \operatorname{cosec}^{-1} x$$



S.No	$f(x)$	Domain	Range
(1)	$\sin^{-1}x$	$ x \leq 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(2)	$\cos^{-1}x$	$ x \leq 1$	$[0, \pi]$
(3)	$\tan^{-1}x$	$x \in \mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
(4)	$\sec^{-1}x$	$ x \geq 1$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$ or $\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
(5)	$\operatorname{cosec}^{-1}x$	$ x \geq 1$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
(6)	$\cot^{-1}x$	$x \in \mathbb{R}$	$(0, \pi)$

From the above discussions following IMPORTANT points can be concluded:

- All the inverse trigonometric functions represent an angle.
- If $x > 0$, then all six inverse trigonometric functions viz $\sin^{-1}x, \cos^{-1}x, \tan^{-1}x, \sec^{-1}x, \operatorname{cosec}^{-1}x, \cot^{-1}x$ represent an acute angle.
- If $x < 0$, then $\sin^{-1}x, \tan^{-1}x$ & $\operatorname{cosec}^{-1}x$ represent an angle from $-\pi/2$ to 0 (IVth quadrant)
- If $x < 0$, then $\cos^{-1}x, \cot^{-1}x$ & $\sec^{-1}x$ represent an obtuse angle. (IInd quadrant)
- IIIrd quadrant is never used in range of inverse trigonometric function.

Illustration 3

Domain of the function $y = \sqrt{\sin^{-1}(\log_2 x)}$

- (A) $[1, 2]$ (B) $[-1, 2]$ (C) $[-2, 2]$ (D) $[-3, 3]$

Ans. (A)

Solution:

$$\sin^{-1}(\log_2 x) \geq 0 \text{ \& } -1 \leq \log_2 x \leq 1 \text{ \& } x > 0$$

$$0 \leq \sin^{-1}(\log_2 x) \leq \frac{\pi}{2}$$

$$0 \leq (\log_2 x) \leq 1$$

$$1 \leq x \leq 2$$

$$D = [1, 2]$$

Illustration 4

Domain of the function $f(x) = \sin^{-1}\left(\frac{2-3[x]}{4}\right)$ (Where $[.]$ is G.I.F.)

- (A) $[-3, 3]$ (B) $[0, 3]$ (C) $[0, 3)$ (D) $(-\infty, 3]$

Ans. (C)

Solution:

$$-1 \leq \frac{2-3[x]}{4} \leq 1 \Rightarrow -4 \leq 2-3[x] \leq 4 \Rightarrow -6 \leq -3[x] \leq 2 \Rightarrow -\frac{2}{3} \leq [x] \leq 2$$

$$[x] = 0, 1, 2 \quad \therefore x \in [0, 3)$$

Inverse Trigonometric Functions

Illustration 5

Domain of the function $y = \sin^{-1}\left(\frac{x-3}{2}\right) + \log_{10}(4-x)$ is

- (A) $[1, 4]$ (B) $[-4, 4]$ (C) $[1, 4)$ (D) $(0, 4)$

Ans. (C)

Solution:

$$y = \sin^{-1}\left(\frac{x-3}{2}\right) + \log_{10}(4-x)$$

$$-1 \leq \frac{x-3}{2} \leq 1 \quad 4-x > 0$$

$$-2 \leq x-3 \leq 2 \quad x < 4$$

$$1 \leq x \leq 5$$

$$D_1 = [1, 5] \quad D_2 = (-\infty, 4) \quad \therefore D = D_1 \cap D_2$$

$$D = [1, 4)$$

Illustration 6

Domain of the function $f(x) = \sin^{-1}(|x-1| - 2)$ is

- (A) $[-2, 0] \cup [2, 4]$ (B) $[-2, -1] \cup [1, 4]$ (C) $(-2, 4)$ (D) $[-2, 4]$

Ans. (A)

Solution:

$$-1 \leq |x-1| - 2 \leq 1$$

$$1 \leq |x-1| \leq 3$$

$$x-1 \in [-3, -1] \cup [1, 3]$$

$$x \in [-2, 0] \cup [2, 4]$$

Illustration 7

Domain of the function $f(x) = \cos^{-1}\left(\frac{6-3x}{4}\right) + \operatorname{cosec}^{-1}\left(\frac{x-1}{2}\right)$ is

- (A) $[3, \infty)$ (B) $\left[3, \frac{10}{3}\right]$ (C) $\left[-3, \frac{10}{3}\right]$ (D) $\left[0, \frac{10}{3}\right]$

Ans. (B)

Solution:

$$-1 \leq \frac{6-3x}{4} \leq 1$$

$$-4 \leq 6-3x \leq 4$$

$$-10 \leq -3x \leq -2$$

$$2 \leq 3x \leq 10$$

$$\frac{2}{3} \leq x \leq \frac{10}{3}$$

$$D_1 = \left[\frac{2}{3}, \frac{10}{3}\right] \text{ and } \left|\frac{x-1}{2}\right| \geq 1$$

$$\frac{x-1}{2} \leq -1 \text{ or } \frac{x-1}{2} \geq 1$$

$$x-1 \leq -2 \text{ or } x-1 \geq 2$$

$$x \leq -1 \text{ or } x \geq 3$$

$$D_2 = (-\infty, -1] \cup [3, \infty)$$

$$\therefore D = D_1 \cap D_2$$

$$= \left[3, \frac{10}{3} \right]$$

Illustration 8

Range of $\sin^{-1}x + \operatorname{cosec}^{-1}x$ is -

- (A) $\{-\pi, \pi\}$ (B) $(-\pi, \pi)$ (C) $\left\{0, \frac{\pi}{2}\right\}$ (D) none of these

Ans. (A)

Solution:

Domain of $\sin^{-1}x$ is $[-1, 1]$

domain of $\operatorname{cosec}^{-1}x$ is $(-\infty, -1] \cup [1, \infty)$

common domain is $\{-1, 1\}$ only

$$\text{range} = \sin^{-1}(-1) + \operatorname{cosec}^{-1}(-1) = -\pi$$

$$\text{or } \sin^{-1}(1) + \operatorname{cosec}^{-1}(1) = \pi$$

$$\text{Hence, range} = \{-\pi, \pi\}$$

Illustration 9

Domain of $f(x) = \sin^{-1}(2x - 2) + \sqrt{\log_{\{x\}} x}$ is, (where $\{.\}$ represents fractional part function):-

- (A) $\left[\frac{1}{2}, \frac{3}{2}\right]$ (B) $\left[\frac{1}{2}, 1\right)$ (C) $\left[\frac{1}{2}, 1\right]$ (D) $\left(\frac{1}{2}, 1\right]$

Ans. (B)

Solution:

$$-1 \leq 2x - 2 \leq 1 \text{ \& } 0 < x < 1$$

$$\frac{1}{2} \leq x \leq \frac{3}{2} \text{ \& } 0 < x < 1$$

$$\text{Intersection } \frac{1}{2} \leq x < 1$$

Illustration 10

Find domain & range of $y = \tan^{-1}(\sqrt{x})$

Solution:

$$D_f : [0, \infty)$$

$$R_f : \left[\tan^{-1}(\sqrt{x}), \right. \\ \left. [0, \infty) \right] \\ \left[0, \frac{\pi}{2} \right)$$

$$R_f : \left[0, \frac{\pi}{2} \right)$$

Illustration 11

Find domain & range of $y = \sin^{-1}(x) + \cos^{-1}(x) + \tan^{-1}(x)$

Solution:

$$D_f : x \in [-1, 1] \text{ \& } x \in [-1, 1] \text{ \& } x \in R$$

$$\Rightarrow x \in [-1, 1]$$

$$\therefore D_f : [-1, 1]$$

$$R_f : \begin{array}{c} \frac{\pi}{2} + \tan^{-1}(x) \\ \text{---} \\ (-1, 1) \\ \text{---} \\ -\frac{\pi}{4}, \frac{\pi}{4} \\ \text{---} \\ \frac{\pi}{4}, \frac{3\pi}{4} \end{array}$$

$$\therefore R_f : \left[\frac{\pi}{4}, \frac{3\pi}{4} \right]$$

Illustration 12

The range of the function $f(x) = \sin^{-1} \left[x^2 + \frac{1}{2} \right] + \cos^{-1} \left[x^2 - \frac{1}{2} \right]$, where $[.]$ is the greatest integer function, is:

Ans. $\{ \pi \}$

Solution:

$$f(x) = \sin^{-1} \left[x^2 + \frac{1}{2} \right] + \cos^{-1} \left[x^2 - \frac{1}{2} \right]$$

$$\text{Domain: } -1 \leq \left[x^2 - \frac{1}{2} \right] \leq 1 \Rightarrow x \in \left(-\sqrt{\frac{5}{2}}, \sqrt{\frac{5}{2}} \right)$$

$$\text{And } -1 \leq \left[x^2 + \frac{1}{2} \right] \leq 1 \Rightarrow x \in \left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}} \right)$$

$$\Rightarrow \text{domain is } x \in \left(-\sqrt{\frac{3}{2}}, \sqrt{\frac{3}{2}} \right) \text{ or } x^2 \in \left[0, \frac{3}{2} \right)$$

$$\text{If (i) } x^2 \in \left[0, \frac{1}{2} \right), \text{ then } f(x) = \pi$$

$$\text{If (ii) } x^2 \in \left[\frac{1}{2}, 1 \right), \text{ then } f(x) = \pi$$

$$\text{If (iii) } x^2 \in \left[1, \frac{3}{2} \right), \text{ then } f(x) = \pi \Rightarrow \text{range} = \{ \pi \}$$

Illustration 13

The domain of the function $f(x) = \sqrt{\frac{1}{(|x|-1)\cos^{-1}(2x+1).\tan 3x}}$ is:

- (A) $(-1, 0)$ (B) $(-1, 0) - \left\{-\frac{\pi}{6}\right\}$
 (C) $(-1, 0) - \left\{-\frac{\pi}{6}, -\frac{\pi}{2}\right\}$ (D) $\left(-\frac{\pi}{6}, 0\right)$

Ans. (D)

Solution:

$$f(x) = \sqrt{\frac{1}{(|x|-1)\cos^{-1}(2x+1).\tan 3x}}$$

$$\text{Here } -1 \leq 2x+1 < 1 \Rightarrow -2 \leq 2x < 0 \Rightarrow -1 \leq x < 0$$

$$\Rightarrow x \in [-1, 0)$$

$$\text{But } x \neq -1 \text{ as } |x| - 1 \neq 0$$

$$\therefore x \in (-1, 0)$$

$$\text{for } x \in (-1, 0), (|x| - 1) \text{ is -ve}$$

$$\therefore \tan 3x < 0$$

$$0 > 3x > -\frac{\pi}{2} \text{ or } x \in \left(-\frac{\pi}{6}, 0\right)$$

$$\text{Domain: } \left(-\frac{\pi}{6}, 0\right) \cap (-1, 0) \equiv \left(-\frac{\pi}{6}, 0\right)$$

Illustration 14

Let $D \equiv [-1, 1]$ is the domain of the following functions, state which of them are injective.

$$(A) f(x) = \begin{cases} \tan^{-1} \frac{1}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$$

$$(B) g(x) = x^3$$

$$(C) h(x) = \sin 2x$$

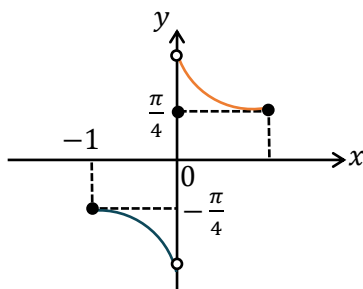
$$(D) k(x) = \sin (\pi x/2)$$

Ans. (B,D)

Solution:

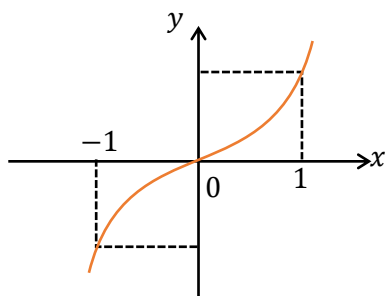
$$\text{Domain } D \in [-1, 1]$$

$$(A) f(x) = \begin{cases} \tan^{-1} \frac{1}{x} & x \neq 0 \\ 1 & x = 0 \end{cases} \quad \text{many-one}$$



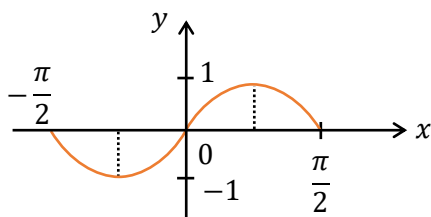
(B) $g(x) = x^3$

one - one



(C) $h(x) = \sin 2x$

many - one



(D) $k(x) = \sin \left(\frac{\pi x}{2} \right)$

one-one function

Illustration 15

Consider $f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$ (where $[.]$ denotes greatest integer function.) If domain of $f(x)$ is $[a, b]$ & the range of $f(x)$ is $\{c, d\}$ then $a + b + \frac{2d}{c}$ is equal to (where $c < d$)

Ans. (4)

Solution:

$$f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$$

$$-1 \leq [2x] \leq 1 \text{ \& } -1 \leq [x] - 1 \leq 1$$

$$\Rightarrow -1 \leq 2x < 2 \text{ \& } 0 \leq [x] \leq 2$$

$$\Rightarrow -\frac{1}{2} \leq x < 1 \text{ \& } 0 \leq x < 3$$

$$\Rightarrow 0 \leq x \leq 1 \text{ Domain}$$

$$\Rightarrow [x] = 0 \Rightarrow 0 \leq 2x < 2$$

$$\Rightarrow [2x] = 0 \text{ or } 1$$

$$\text{Now } f(x) = \sin^{-1}[2x] + \cos^{-1}(-1)$$

$$= \left(0 \text{ or } \frac{\pi}{2}\right) + \pi = \pi \text{ or } \frac{3\pi}{2}$$

$$\Rightarrow a + b + \frac{2d}{c} = 1 + 3 = 4$$

Illustration 16

The value of $\cos \left(\sin^{-1} \left(\sqrt{\frac{3-\sqrt{5}}{8}} \right) + 2 \cos^{-1} \left(\sqrt{\frac{3+\sqrt{5}}{8}} \right) \right)$ is equal to -

(A) 1

(B) -1

(C) $\frac{1}{2}$

(D) 0

Ans. (D)

Solution:

$$\sqrt{\frac{3-\sqrt{5}}{8}} = \sqrt{\frac{6-2\sqrt{5}}{16}} = \frac{(\sqrt{5}-1)}{4}$$

$$\sin^{-1}\left(\frac{\sqrt{5}-1}{4}\right) = \frac{\pi}{10}$$

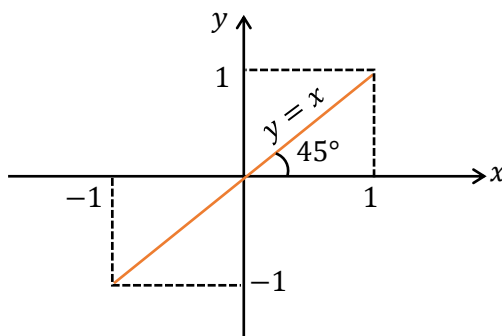
$$\text{Similarly, } \cos^{-1}\sqrt{\frac{3+\sqrt{5}}{8}} = \frac{\pi}{5}$$

$$\cos\left(\frac{\pi}{10} + \frac{2\pi}{5}\right) = \cos\left(\frac{5\pi}{10}\right) = 0$$

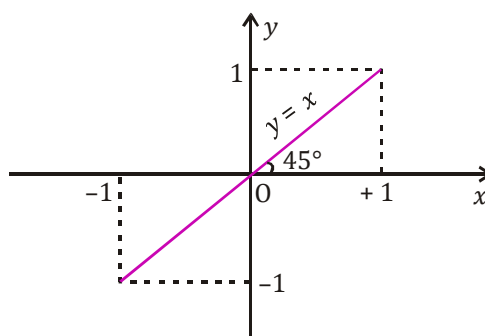
3. PROPERTIES OF INVERSE CIRCULAR FUNCTIONS:

3.1 Property -1

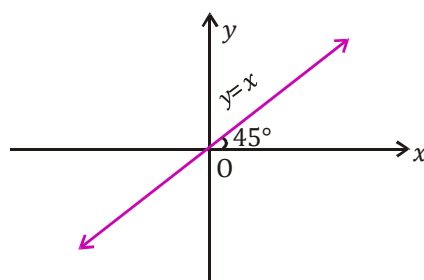
(i) $y = \sin(\sin^{-1}x) = x$
 $x \in [-1,1], y \in [-1,1]$



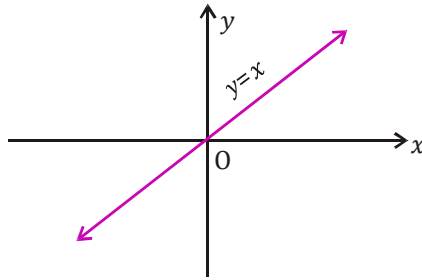
(ii) $y = \cos(\cos^{-1}x) = x$
 $x \in [-1,1], y \in [-1,1]$



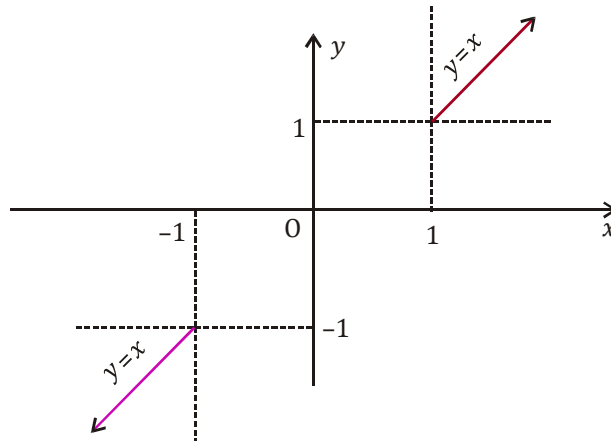
(iii) $y = \tan(\tan^{-1}x) = x$
 $x \in \mathbb{R}, y \in \mathbb{R}$



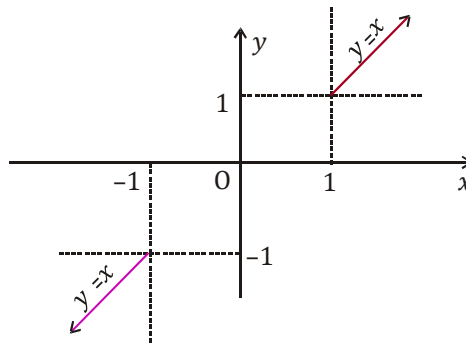
(iv) $y = \cot(\cot^{-1} x) = x$,
 $x \in \mathbb{R}; y \in \mathbb{R}$



(v) $y = \operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$
 $|x| \geq 1, |y| \geq 1$



(vi) $y = \sec(\sec^{-1} x) = x$
 $|x| \geq 1; |y| \geq 1$



Note: All the above functions are aperiodic.

Illustration 17

Evaluate the following:

(i) $\sin\left(\cos^{-1} \frac{3}{5}\right)$ (ii) $\cos\left(\tan^{-1} \frac{3}{4}\right)$

Solution:

(i) Let $\cos^{-1} 3/5 = \theta$. Then,

$$\cos \theta = 3/5 \Rightarrow \sin \theta = 4/5 \therefore \sin(\cos^{-1} 3/5) = \sin \theta = 4/5$$

(ii) Let $\tan^{-1} 3/4 = \theta$. Then,

$$\tan \theta = 3/4$$

$$\Rightarrow \cos \theta = \frac{4}{5} \left\{ \because \text{as } \cos^2 \theta = \frac{1}{1 + \tan^2 \theta} \right\} \therefore \cos(\tan^{-1} 3/4) = \cos \theta = 4/5$$

Illustration 18

$$(1) \sin\left(\frac{\pi}{2} - \sin^{-1}\left(-\frac{1}{2}\right)\right) \quad (2) \tan\left\{2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right\}$$

Solution:

$$(1) \sin\left(\frac{\pi}{2} - \sin^{-1}\left(-\frac{1}{2}\right)\right) = \sin\left(\frac{\pi}{2} - \left(-\frac{\pi}{6}\right)\right) = \sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$$

$$(2) \text{ Let } \tan^{-1}\left(\frac{1}{5}\right) = \theta \Rightarrow \tan \theta = \frac{1}{5}$$

$$\tan\left(2\theta - \frac{\pi}{4}\right) = \frac{\tan(2\theta) - 1}{1 + \tan(2\theta)} \text{ and } \tan 2\theta = \frac{2 \times \frac{1}{5}}{1 - \frac{1}{25}} = \frac{5}{12} \quad (\because \tan \theta = \frac{1}{5})$$

$$\Rightarrow \tan\left(2\theta - \frac{\pi}{4}\right) = \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} = -\frac{7}{17}$$

Illustration 19

Prove that $\sec^2(\tan^{-1} 2) + \operatorname{cosec}^2(\cot^{-1} 3) = 15$

Solution:

We have,

$$\sec^2(\tan^{-1} 2) + \operatorname{cosec}^2(\cot^{-1} 3)$$

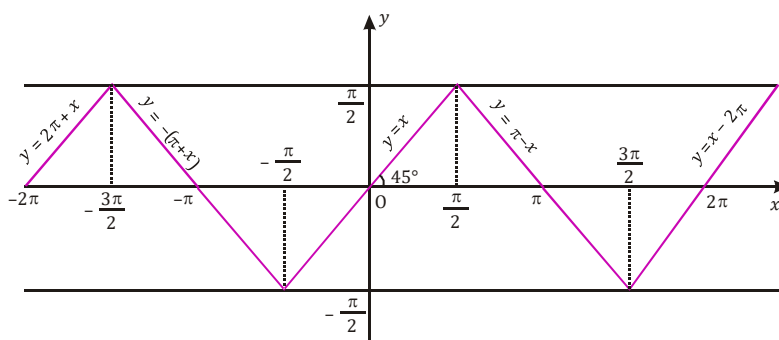
$$= \left\{\sec(\tan^{-1} 2)\right\}^2 + \left\{\operatorname{cosec}(\cot^{-1} 3)\right\}^2 = \left\{\sec\left(\tan^{-1} \frac{2}{1}\right)\right\}^2 + \left\{\operatorname{cosec}\left(\cot^{-1} \frac{3}{1}\right)\right\}^2$$

$$= \left\{\sec(\sec^{-1} \sqrt{5})\right\}^2 + \left\{\operatorname{cosec}(\operatorname{cosec}^{-1} \sqrt{10})\right\}^2 = (\sqrt{5})^2 + (\sqrt{10})^2 = 15$$

3.2 Property -2

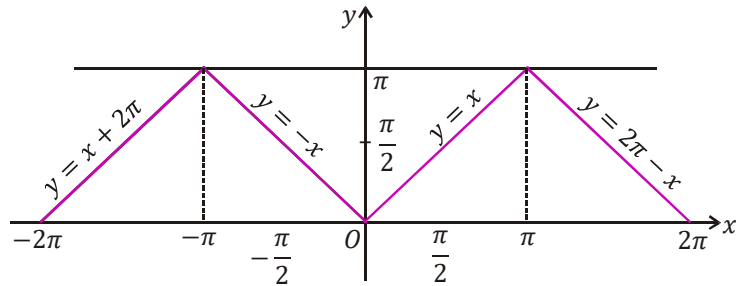
(i) $y = \sin^{-1}(\sin x), x \in \mathbb{R}, y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ periodic with period 2π and it is an odd function.

$$\sin^{-1}(\sin x) = \begin{cases} -\pi - x, & -\pi \leq x \leq -\frac{\pi}{2} \\ x, & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} \leq x \leq \pi \end{cases}$$



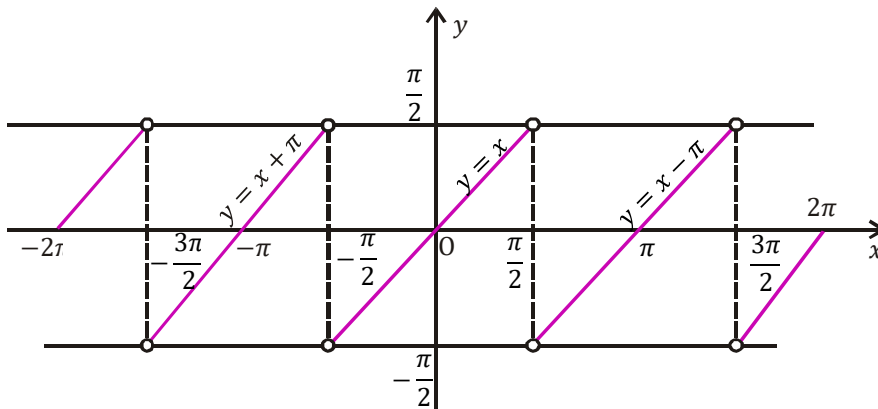
(ii) $y = \cos^{-1}(\cos x)$, $x \in \mathbb{R}$, $y \in [0, \pi]$, periodic with period 2π and it is an even function.

$$\cos^{-1}(\cos x) = \begin{cases} -x & , \quad -\pi \leq x \leq 0 \\ x & , \quad 0 < x \leq \pi \end{cases}$$



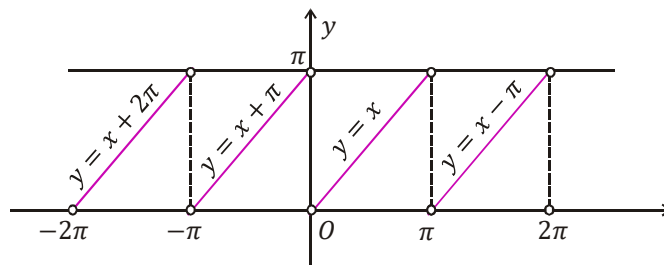
(iii) $y = \tan^{-1}(\tan x)$ $x \in \mathbb{R} - \left\{ (2n-1)\frac{\pi}{2}, n \in \mathbb{I} \right\}$; $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$, periodic with period π and it is an odd function.

$$\tan^{-1}(\tan x) = \begin{cases} x + \pi & , \quad -\frac{3\pi}{2} < x < -\frac{\pi}{2} \\ x & , \quad -\frac{\pi}{2} < x < \frac{\pi}{2} \\ x - \pi & , \quad \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

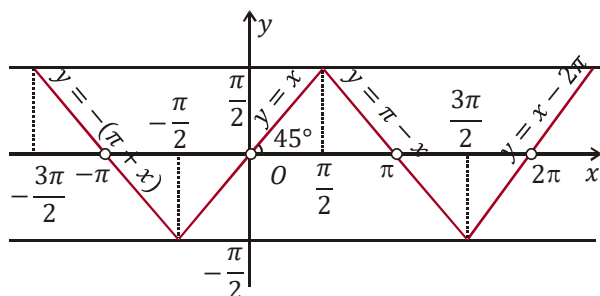


(iv) $y = \cot^{-1}(\cot x)$, $x \in \mathbb{R} - \{n\pi, n \in \mathbb{I}\}$, $y \in (0, \pi)$, periodic with period π and neither even nor odd function.

$$\cot^{-1}(\cot x) = \begin{cases} x + \pi & , \quad -\pi < x < 0 \\ x & , \quad 0 < x < \pi \\ x - \pi & , \quad \pi < x < 2\pi \end{cases}$$



- (v) $y = \operatorname{cosec}^{-1}(\operatorname{cosec} x)$, $x \in \mathbb{R} - \{n\pi, n \in I\}$ $y \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$, is periodic with period 2π and it is an odd function.



- (vi) $y = \sec^{-1}(\sec x)$, y is periodic with period 2π and it is an even function.

$$x \in \mathbb{R} - \left\{(2n-1)\frac{\pi}{2}, n \in I\right\}, y \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$$

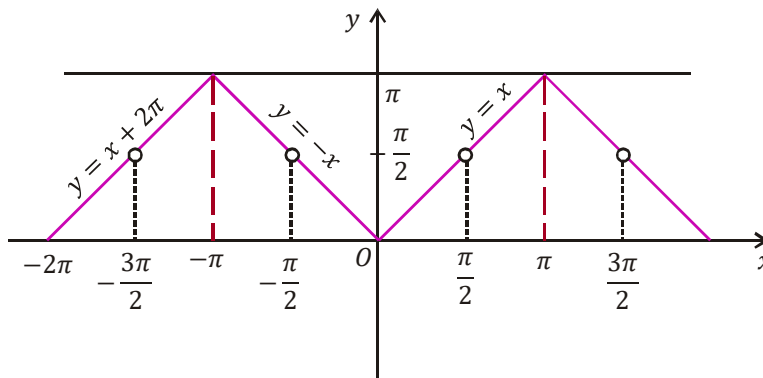


Illustration 20

The value of $\sin^{-1}(-\sqrt{3}/2) + \cos^{-1}(\cos(7\pi/6))$ is -

- (A) $5\pi/6$ (B) $\pi/2$ (C) $3\pi/2$ (D) none of these

Ans. (B)

Solution:

$$\sin^{-1}\left(-\sqrt{3}/2\right) = -\sin^{-1}\left(\sqrt{3}/2\right) = -\pi/3$$

$$\text{and } \cos^{-1}(\cos(7\pi/6)) = \cos^{-1}(\cos(2\pi - 5\pi/6)) = \cos^{-1}(\cos(5\pi/6)) = 5\pi/6$$

$$\text{Hence } \sin^{-1}(-\sqrt{3}/2) + \cos^{-1}(\cos(7\pi/6)) = -\frac{\pi}{3} + \frac{5\pi}{6} = \frac{\pi}{2}$$

Illustration 21

Evaluate the following:

- (i) $\sin^{-1}(\sin 10)$ (ii) $\tan^{-1}(\tan(-6))$

Solution:

- (i) We know that $\sin^{-1}(\sin \theta) = \theta$, if $-\pi/2 \leq \theta \leq \pi/2$

Here, $\theta = 10$ radians which does not lie between $-\pi/2$ and $\pi/2$

But, $3\pi - \theta$ i.e., $3\pi - 10$ lie between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

$$\text{Also, } \sin(3\pi - 10) = \sin 10$$

$$\therefore \sin^{-1}(\sin 10) = \sin^{-1}(\sin(3\pi - 10)) = (3\pi - 10)$$

Inverse Trigonometric Functions

(ii) We know that,

$\tan^{-1}(\tan\theta) = \theta$, if $-\pi/2 < \theta < \pi/2$. Here, $\theta = -6$, radians which does not lie between $-\pi/2$ and $\pi/2$.

We find that $2\pi - 6$ lies between $-\pi/2$ and $\pi/2$ such that;

$$\tan(2\pi - 6) = -\tan 6 = \tan(-6)$$

$$\therefore \tan^{-1}(\tan(-6)) = \tan^{-1}(\tan(2\pi - 6)) = (2\pi - 6)$$

Illustration 22

Evaluate the following:

$$\cot^{-1}(\cot 4)$$

Solution:

$$\cot^{-1}(\cot 4) = \cot^{-1}(\cot(\pi + (4 - \pi))) = \cot^{-1}(\cot(4 - \pi)) = (4 - \pi)$$

Illustration 23

The principle value of $\cos^{-1}\left(-\sin\frac{7\pi}{6}\right)$ is -

(A) $\frac{5\pi}{3}$

(B) $\frac{7\pi}{6}$

(C) $\frac{\pi}{3}$

(D) none of these

Solution:

Ans. (C)

$$\frac{\pi}{2} - \sin^{-1}\left(-\sin\left(\frac{7\pi}{6}\right)\right) = \frac{\pi}{2} + \sin^{-1}\sin\left(\frac{7\pi}{6}\right) = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$$

Illustration 24

$\sin^{-1}\left(\sin\frac{4\pi}{7}\right) + \cos^{-1}\left(\cos\frac{4\pi}{7}\right) + \tan^{-1}\left(\tan\frac{4\pi}{7}\right)$ is equal to -

(A) π

(B) $\frac{8\pi}{7}$

(C) $\frac{4\pi}{7}$

(D) $\frac{12\pi}{7}$

Ans. (C)

Solution:

$$\therefore \frac{\pi}{2} < \frac{4\pi}{7} < \pi$$

$$\therefore \sin^{-1}\left\{\sin\left(\pi - \frac{4\pi}{7}\right)\right\} + \cos^{-1}\left(\cos\frac{4\pi}{7}\right) + \tan^{-1}\left\{\tan\left(\frac{4\pi}{7} - \pi\right)\right\}$$

$$\cos^{-1}\left(\cos\frac{4\pi}{7}\right) = \frac{4\pi}{7}$$

Illustration 25

Identify the correct statement(s) -

(A) $\tan^{-1}\tan\left(\frac{47}{7}\pi\right)$ is negative

(B) $\cos^{-1}(\cos(1 + \sin x)) = 1 + \sin x$ for all $x \in R$

(C) $\sin^{-1}\left(\sin\frac{23\pi}{5}\right) + \cos^{-1}\left(\cos\frac{23\pi}{5}\right) = \pi$

(D) $\tan^{-1} 2 > \cot^{-1} 2$

Ans. (A,B,C,D)

Solution:

$$(A) \tan^{-1} \left\{ \tan \left(6\pi - \frac{2\pi}{7} \right) \right\} = -\frac{2\pi}{7}$$

$$(B) 1 + \sin x \in [0, 2], \forall x \in \mathbb{R}$$

$$\Rightarrow \cos^{-1}(\cos(1 + \sin x)) = 1 + \sin x$$

$$(C) \sin^{-1} \left(\sin \frac{23\pi}{5} \right) = \sin^{-1} \left(\sin \left(5\pi - \frac{2\pi}{5} \right) \right) = \frac{2\pi}{5}$$

$$\cos^{-1} \left(\cos \frac{23\pi}{5} \right) = \cos^{-1} \left(\cos \left(4\pi + \frac{3\pi}{5} \right) \right) = \frac{3\pi}{5}$$

$$\therefore \frac{2\pi}{5} + \frac{3\pi}{5} = \pi$$

$$(D) \tan^{-1} 2 > \frac{\pi}{2} - \tan^{-1} 2 \Leftrightarrow \tan^{-1} 2 > \frac{\pi}{4} \Leftrightarrow \tan^{-1} 2 > \tan^{-1} 1 \text{ which is true}$$

3.3 Property -3

$$(i) \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}; \quad |x| \leq 1$$

$$(ii) \tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}; \quad x \in \mathbb{R}$$

$$(iii) \operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}; \quad |x| \geq 1$$

3.4 Property -4

$$(i) \sin^{-1}(-x) = -\sin^{-1} x; \quad |x| \leq 1$$

$$(ii) \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x; \quad |x| \geq 1$$

$$(iii) \tan^{-1}(-x) = -\tan^{-1} x; \quad x \in \mathbb{R}$$

$$(iv) \cot^{-1}(-x) = \pi - \cot^{-1} x; \quad x \in \mathbb{R}$$

$$(v) \cos^{-1}(-x) = \pi - \cos^{-1} x; \quad |x| \leq 1$$

$$(vi) \sec^{-1}(-x) = \pi - \sec^{-1} x; \quad |x| \geq 1$$

Illustration 26

Let $y = \cos^{-1} x + \cot^{-1} x$, then y can be equal to -

$$(A) \frac{\pi}{6} \quad (B) \frac{\pi}{2} \quad (C) \frac{3\pi}{2} \quad (D) \frac{11\pi}{6}$$

Ans. (B,C)

Solution:

$$y = \cos^{-1} x + \cot^{-1} x$$

Domain of y is $x \in [-1, 1]$

$\therefore$ both $\cos^{-1} x$ & $\cot^{-1} x$ are decreasing

$$\Rightarrow y_{\max} = \cos^{-1}(-1) + \cot^{-1}(-1)$$

$$= \pi + \frac{3\pi}{4} = \frac{7\pi}{4}$$

$$y_{\min} = \cos^{-1}(1) + \cot^{-1}(1) = 0 + \frac{\pi}{4}$$

$$\Rightarrow y \in \left[\frac{\pi}{4}, \frac{7\pi}{4} \right]$$

$\Rightarrow$ (B),(C) are correct.

Inverse Trigonometric Functions

3.5 Property -5

$$(i) \operatorname{cosec}^{-1} x = \sin^{-1} \frac{1}{x}; \quad |x| \geq 1$$

$$(ii) \sec^{-1} x = \cos^{-1} \frac{1}{x}; \quad |x| \geq 1$$

$$(iii) \cot^{-1} x = \begin{cases} \tan^{-1} \frac{1}{x} & ; x > 0 \\ \pi + \tan^{-1} \frac{1}{x} & ; x < 0 \end{cases}$$

3.6 Property -6

$$(i) (a) \tan^{-1} x + \tan^{-1} y = \begin{cases} \tan^{-1} \frac{x+y}{1-xy} & \text{where } x > 0, y > 0 \text{ \& } xy < 1 \\ \pi + \tan^{-1} \frac{x+y}{1-xy} & \text{where } x > 0, y > 0 \text{ \& } xy > 1 \\ \frac{\pi}{2}, & \text{where } x > 0, y > 0 \text{ \& } xy = 1 \end{cases}$$

$$(b) \tan^{-1} x - \tan^{-1} y = \tan^{-1} \frac{x-y}{1+xy} \text{ where } x > 0, y > 0$$

$$(c) \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x+y+z-xyz}{1-xy-yz-zx} \right]$$

$$\text{if } x > 0, y > 0, z > 0 \text{ \& } xy + yz + zx < 1$$

$$(ii) (a) \sin^{-1} x + \sin^{-1} y = \begin{cases} \sin^{-1} [x\sqrt{1-y^2} + y\sqrt{1-x^2}] & \text{where } x > 0, y > 0 \text{ \& } (x^2 + y^2) \leq 1 \\ \pi - \sin^{-1} [x\sqrt{1-y^2} + y\sqrt{1-x^2}] & \text{where } x > 0, y > 0 \text{ \& } x^2 + y^2 > 1 \end{cases}$$

$$(b) \sin^{-1} x - \sin^{-1} y = \sin^{-1} [x\sqrt{1-y^2} - y\sqrt{1-x^2}] \text{ where } x > 0, y > 0$$

$$(iii) (a) \cos^{-1} x + \cos^{-1} y = \cos^{-1} [xy - \sqrt{1-x^2}\sqrt{1-y^2}] \text{ where } x > 0, y > 0$$

$$(b) \cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1} (xy + \sqrt{1-x^2}\sqrt{1-y^2}); & x < y, x, y > 0 \\ -\cos^{-1} (xy + \sqrt{1-x^2}\sqrt{1-y^2}); & x > y, x, y > 0 \end{cases}$$

Note: In the above results x & y are taken positive. In case if these are given as negative, we first apply P-4 and then use above results.

Illustration 27

Prove that $\tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} = \tan^{-1} \frac{2}{9}$

Solution:

$$\text{L. H. S.} = \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13}$$

$$= \tan^{-1} \left\{ \frac{\frac{1}{7} + \frac{1}{13}}{1 - \frac{1}{7} \times \frac{1}{13}} \right\} \quad \left\{ \because \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right); \text{ if } xy < 1 \right\}$$

$$\tan^{-1} \left(\frac{20}{90} \right) = \tan^{-1} \left(\frac{2}{9} \right) = \text{R.H.S.}$$

Illustration 28

Prove that $\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$

Solution:

$$\begin{aligned} & \left(\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} \right) + \left(\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{8} \right) \\ &= \tan^{-1} \left(\frac{\frac{1}{5} + \frac{1}{7}}{1 - \frac{1}{5} \times \frac{1}{7}} \right) + \tan^{-1} \left(\frac{\frac{1}{3} + \frac{1}{8}}{1 - \frac{1}{3} \times \frac{1}{8}} \right) = \tan^{-1} \left(\frac{6}{17} \right) + \tan^{-1} \left(\frac{11}{23} \right) \\ &= \tan^{-1} \left(\frac{\frac{6}{17} + \frac{11}{23}}{1 - \frac{6}{17} \times \frac{11}{23}} \right) = \tan^{-1} \left(\frac{325}{325} \right) = \tan^{-1}(1) = \frac{\pi}{4} \end{aligned}$$

Illustration 29

Prove that $\sin^{-1} \frac{12}{13} + \cot^{-1} \frac{4}{3} + \tan^{-1} \frac{63}{16} = \pi$

Solution:

We have,

$$\begin{aligned} & \sin^{-1} \frac{12}{13} + \cos^{-1} \frac{4}{5} + \tan^{-1} \frac{63}{16} \\ &= \tan^{-1} \frac{12}{5} + \tan^{-1} \frac{3}{4} + \tan^{-1} \frac{63}{16} \quad \left[\because \sin^{-1} \frac{12}{13} = \tan^{-1} \frac{12}{5} \text{ and } \cos^{-1} \frac{4}{5} = \tan^{-1} \frac{3}{4} \right] \\ &= \pi + \tan^{-1} \left\{ \frac{\frac{12}{5} + \frac{3}{4}}{1 - \frac{12}{5} \times \frac{3}{4}} \right\} + \tan^{-1} \frac{63}{16} \quad \left[\because \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1} \left(\frac{x+y}{1-xy} \right), \text{ if } xy > 1 \right] \\ &= \pi + \tan^{-1} \left(\frac{63}{-16} \right) + \tan^{-1} \left(\frac{63}{16} \right) \\ &= \pi - \tan^{-1} \frac{63}{16} + \tan^{-1} \frac{63}{16} = \pi \quad \left[\because \tan^{-1}(-x) = -\tan^{-1} x \right] \end{aligned}$$

Illustration 30

Prove that: $\cos^{-1} \frac{12}{13} + \sin^{-1} \frac{3}{5} = \sin^{-1} \frac{56}{65}$

Solution:

$$\begin{aligned} & \text{We have, L.H.S.} = \cos^{-1} \left(\frac{12}{13} \right) + \sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{5}{12} \right) + \tan^{-1} \left(\frac{3}{4} \right) \\ & \left[\because \cos^{-1} \left(\frac{12}{13} \right) = \tan^{-1} \left(\frac{5}{12} \right) \text{ \& } \sin^{-1} \left(\frac{3}{5} \right) = \tan^{-1} \left(\frac{3}{4} \right) \right] \\ & \text{L.H.S.} = \tan^{-1} \left(\frac{\frac{5}{12} + \frac{3}{4}}{1 - \frac{5}{12} \times \frac{3}{4}} \right) = \tan^{-1} \left(\frac{56}{33} \right) \\ & \text{R.H.S.} = \sin^{-1} \left(\frac{56}{65} \right) = \tan^{-1} \left(\frac{56}{33} \right) \\ & \text{L.H.S.} = \text{R.H.S. Hence Proved} \end{aligned}$$

Illustration 31

Which of the following is true -

- (A) $\cos^{-1} \sqrt{\frac{1+x}{2}} = \sin^{-1} \sqrt{\frac{1-x}{2}} \quad \forall x \in [-1, 1]$ (B) $\cos^{-1} \sqrt{\frac{1+x}{2}} = 2 \cos^{-1} x \quad \forall x \in [-1, 1]$
 (C) $\sin^{-1} \sqrt{\frac{1-x}{2}} = \frac{1}{2} \cos^{-1} x \quad \forall x \in [-1, 1]$ (D) $\sin^{-1} \sqrt{\frac{1-x}{2}} = \frac{1}{2} \sin^{-1} x \quad \forall x \in [-1, 1]$

Ans. (A,C)

Solution:

Let $\cos^{-1} x = \theta \in [0, \pi]$

$$\Rightarrow \cos^{-1} \sqrt{\frac{1+x}{2}} = \cos^{-1} \left(\cos \frac{\theta}{2} \right) = \frac{\theta}{2} = \frac{1}{2} \cos^{-1} x$$

$$\text{also } \sin^{-1} \sqrt{\frac{1-x}{2}} = \sin^{-1} \left(\sin \frac{\theta}{2} \right) = \frac{1}{2} \cos^{-1} x$$

$\therefore$ A, C are correct.

Illustration 32

If $\theta = \tan^{-1}(2\tan^2\theta) - \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5+4 \cos 2\theta} \right)$ then find the sum of all possible values of $\tan \theta$.

Solution:

$$\theta = \tan^{-1}(2\tan^2\theta) - \frac{1}{2} \sin^{-1} \left(\frac{3 \sin 2\theta}{5+4 \cos 2\theta} \right) \Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \frac{1}{2} \sin^{-1} \left(\frac{6 \tan \theta}{9+\tan^2 \theta} \right)$$

$$\Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \frac{1}{2} \sin^{-1} \left[\frac{2 \left(\frac{1}{3} \tan \theta \right)}{1 + \left(\frac{1}{3} \tan \theta \right)^2} \right] \Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \frac{2}{2} \tan^{-1} \left(\frac{1}{3} \tan \theta \right)$$

$$\Rightarrow \theta = \tan^{-1}(2\tan^2\theta) - \tan^{-1} \left(\frac{1}{3} \tan \theta \right) \quad \dots(i)$$

taking tangent on both sides

$$\Rightarrow \tan \theta = \frac{6 \tan^2 \theta - \tan \theta}{3 + 2 \tan^3 \theta} \Rightarrow 2 \tan^4 \theta - 6 \tan^2 \theta + 4 \tan \theta = 0$$

$$\Rightarrow 2 \tan \theta (\tan^3 \theta - 3 \tan \theta + 2) = 0 \Rightarrow 2 \tan \theta (\tan \theta - 1)^2 (\tan \theta + 2) = 0$$

$$\Rightarrow \tan \theta = 0, 1, -2 \text{ which satisfy equation (i)}$$

$$\therefore \text{sum} = 0 + 1 - 2 = -1$$

Illustration 33

If $\sin^{-1} \left(\frac{\sqrt{x}}{2} \right) + \sin^{-1} \left(\sqrt{1 - \frac{x}{4}} \right) + \tan^{-1}(y) = \frac{2\pi}{3}$, then

- (A) Maximum value of $x^2 + y^2$ is $\frac{49}{3}$ (B) Maximum value of $x^2 + y^2$ is 4
 (C) Minimum value of $x^2 + y^2$ is $\frac{1}{3}$ (D) Minimum value of $x^2 + y^2$ is 3

Ans. (A,C)

Solution:

$$\sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right) = \theta \Rightarrow \sin \theta = \sqrt{1-\frac{x}{4}}$$

$$\frac{\pi}{2} + \tan^{-1} y = \frac{2\pi}{3} \Rightarrow y = \frac{1}{\sqrt{3}} \text{ and } 0 \leq x \leq 4$$

$$\text{Max. value of } x^2 + y^2 \text{ is } \frac{49}{3}$$

$$\text{Min. value of } x^2 + y^2 \text{ is } \frac{1}{3}$$

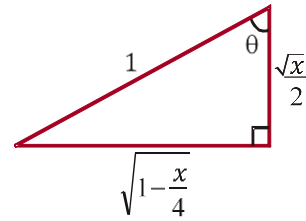


Illustration 34

Let x_1, x_2, x_3 be the solution of $\tan^{-1}\left(\frac{2x+1}{x+1}\right) + \tan^{-1}\left(\frac{2x-1}{x-1}\right) = 2\tan^{-1}(x+1)$ where $x_1 < x_2 < x_3$ then $2x_1 + x_2 + x_3^2$ is equal to

Ans. (1)

Solution:

$$\text{Let } \alpha + \beta = 2\gamma \quad \dots(i)$$

$$\text{where } \tan \alpha = \frac{2x+1}{x+1}, \tan \beta = \frac{2x-1}{x-1} \text{ \& } \tan \gamma = x+1$$

Taking tan in (i) we get

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} = \frac{2 \tan \gamma}{1 - \tan^2 \gamma}$$

$$\Rightarrow \frac{\frac{2x+1}{x+1} + \frac{2x-1}{x-1}}{1 - \frac{4x^2-1}{x^2-1}} = \frac{2(x+1)}{1 - (x+1)^2} \Rightarrow \frac{2x^2 - x - 1 + 2x^2 + x - 1}{x^2 - 1 - 4x^2 + 1} = \frac{2(x+1)}{-x^2 - 2x}$$

$$\Rightarrow \frac{2(2x^2 - 1)}{-3x^2} = \frac{2(x+1)}{-(x^2 + 2x)}$$

$$\Rightarrow (2x^2 - 1)x(x+2) = 3(x+1)x$$

$$\Rightarrow x(2x^3 + 4x^2 - x - 2) = 3x^2(x+1)$$

$$x = 0 \text{ Or } 2x^3 + 4x^2 - x - 2 = 3x^2 + 3x$$

$$\Rightarrow 2x^3 + x^2 - 4x - 2 = 0 \Rightarrow x^2(2x+1) - 2(2x+1) = 0$$

$$\Rightarrow (x^2 - 2)(2x+1) = 0 \Rightarrow x = \sqrt{2}, -\sqrt{2} \text{ or } x = -\frac{1}{2}$$

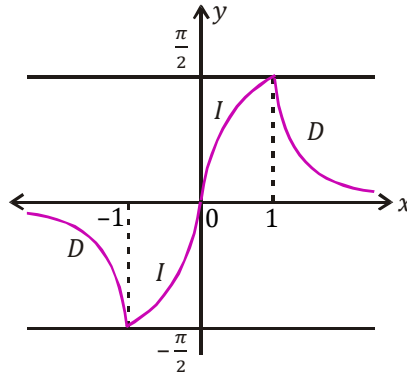
$x = -\sqrt{2}$ is rejected

$\therefore$ it does not satisfy (i)

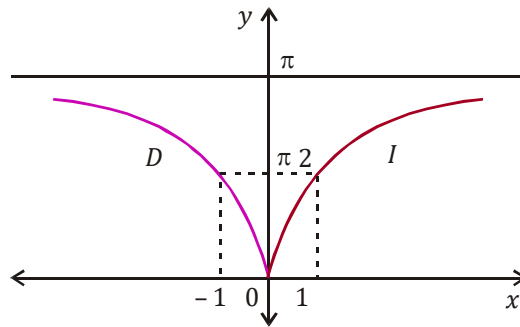
$$\Rightarrow x_1 = -\frac{1}{2}, x_2 = 0 \text{ \& } x_3 = \sqrt{2} \Rightarrow 2x_1 + x_2 + x_3^2 = 1$$

4. SIMPLIFIED INVERSE TRIGONOMETRIC FUNCTIONS:

$$(a) \quad y = f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1} x & \text{if } |x| \leq 1 \\ \pi - 2\tan^{-1} x & \text{if } x > 1 \\ -(\pi + 2\tan^{-1} x) & \text{if } x < -1 \end{cases}$$



$$(b) \quad y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1} x & \text{if } x \geq 0 \\ -2\tan^{-1} x & \text{if } x < 0 \end{cases}$$



$$(c) \quad y = f(x) = \tan^{-1}\frac{2x}{1-x^2} = \begin{cases} 2\tan^{-1} x & \text{if } |x| < 1 \\ \pi + 2\tan^{-1} x & \text{if } x < -1 \\ -(\pi - 2\tan^{-1} x) & \text{if } x > 1 \end{cases}$$

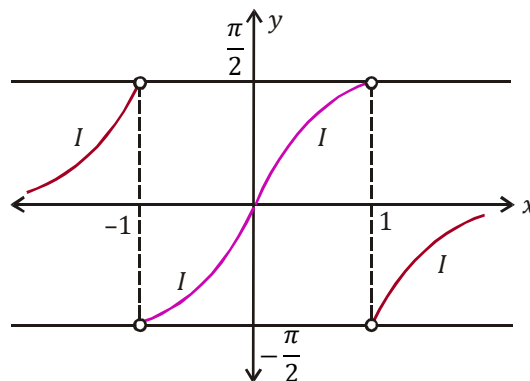


Illustration 35

Prove that: $2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7} = \tan^{-1} \frac{31}{17}$

Solution:

We have, $2 \tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{7}$

$$= \tan^{-1} \left\{ \frac{2 \times \frac{1}{2}}{1 - \left(\frac{1}{2}\right)^2} \right\} + \tan^{-1} \frac{1}{7} \quad \left[\because 2 \tan^{-1} x = \tan^{-1} \left(\frac{2x}{1-x^2} \right), \text{ if } -1 < x < 1 \right]$$

$$= \tan^{-1} \frac{4}{3} + \tan^{-1} \frac{1}{7} = \tan^{-1} \left\{ \frac{\frac{4}{3} + \frac{1}{7}}{1 - \frac{4}{3} \times \frac{1}{7}} \right\} = \tan^{-1} \frac{31}{17}$$

Illustration 36

Prove that $\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right), x \in [0, 1]$

Solution:

$$\text{We have, } \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) = \frac{1}{2} \cos^{-1} \left\{ \frac{1 - (\sqrt{x})^2}{1 + (\sqrt{x})^2} \right\} = \frac{1}{2} \times 2 \tan^{-1} \sqrt{x} = \tan^{-1} \sqrt{x}.$$

Alter: Putting $\sqrt{x} = \tan \theta$, we have $\Rightarrow \theta \in \left[0, \frac{\pi}{4} \right]$

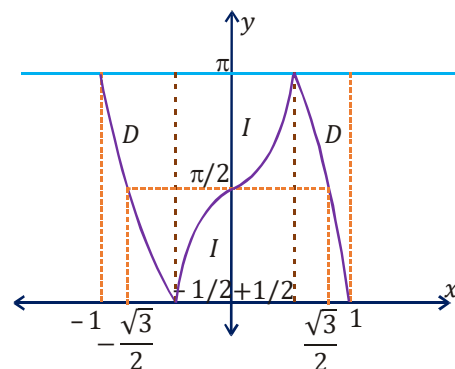
$$\text{RHS} = \frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right) = \frac{1}{2} \cos^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right) = \frac{1}{2} \cos^{-1} (\cos 2\theta) = \theta \quad \because \left(2\theta \in \left[0, \frac{\pi}{2} \right] \right)$$

$$= \tan^{-1} \sqrt{x} = \text{LHS}$$

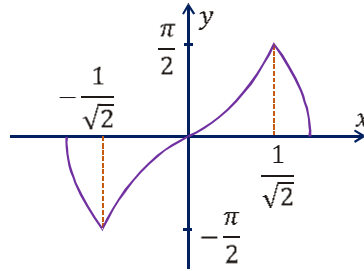
$$(d) \quad y = f(x) = \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3\sin^{-1} x) & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 3\sin^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3\sin^{-1} x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$(e) \quad y = f(x) = \cos^{-1}(4x^3 - 3x)$$

$$= \begin{cases} 3\cos^{-1} x - 2\pi & \text{if } -1 \leq x \leq -\frac{1}{2} \\ 2\pi - 3\cos^{-1} x & \text{if } -\frac{1}{2} \leq x \leq \frac{1}{2} \\ 3\cos^{-1} x & \text{if } \frac{1}{2} \leq x \leq 1 \end{cases}$$



$$(f) \quad \sin^{-1}(2x\sqrt{1-x^2}) = \begin{cases} -(\pi + 2\sin^{-1} x) & -1 \leq x \leq -\frac{1}{\sqrt{2}} \\ 2\sin^{-1} x & -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}} \\ \pi - 2\sin^{-1} x & \frac{1}{\sqrt{2}} \leq x \leq 1 \end{cases}$$



$$(g) \quad \cos^{-1}(2x^2 - 1) = \begin{cases} 2\cos^{-1} x & 0 \leq x \leq 1 \\ 2\pi - 2\cos^{-1} x & -1 \leq x \leq 0 \end{cases}$$

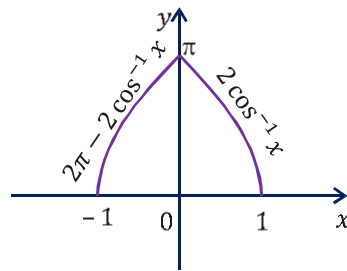


Illustration 37

Prove that: $4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99} = \frac{\pi}{4}$

Solution:

$$\begin{aligned} 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99} &= 2\left\{2\tan^{-1}\frac{1}{5}\right\} - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99} \\ &= 2\left\{\tan^{-1}\frac{2 \times 1/5}{1 - (1/5)^2}\right\} - \tan^{-1}\frac{1}{70} + \tan^{-1}\frac{1}{99} \left[\begin{array}{l} \because 2\tan^{-1} x \\ = \tan^{-1} \frac{2x}{1-x^2}, \text{ if } |x| < 1 \end{array} \right] \\ &= 2\tan^{-1}\frac{5}{12} - \left\{\tan^{-1}\frac{1}{70} - \tan^{-1}\frac{1}{99}\right\} = \tan^{-1}\left\{\frac{2 \times 5/12}{1 - (5/12)^2}\right\} - \tan^{-1}\left\{\frac{\frac{1}{70} - \frac{1}{99}}{1 + \frac{1}{70} \times \frac{1}{99}}\right\} \\ &= \tan^{-1}\frac{120}{119} - \tan^{-1}\frac{29}{6931} = \tan^{-1}\frac{120}{119} - \tan^{-1}\frac{1}{239} = \tan^{-1}\left\{\frac{\frac{120}{119} - \frac{1}{239}}{1 + \frac{120}{119} \times \frac{1}{239}}\right\} = \tan^{-1}1 = \frac{\pi}{4} \end{aligned}$$

Illustration 38

Prove that: $2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2\tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

Solution:

$$\begin{aligned} 2\tan^{-1}\frac{1}{5} + \sec^{-1}\frac{5\sqrt{2}}{7} + 2\tan^{-1}\frac{1}{8} &= 2\left\{\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8}\right\} + \sec^{-1}\frac{5\sqrt{2}}{7} \\ &= 2\tan^{-1}\left\{\frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \times \frac{1}{8}}\right\} + \tan^{-1}\sqrt{\left(\frac{5\sqrt{2}}{7}\right)^2 - 1} \quad \left[\because \sec^{-1}x = \tan^{-1}\sqrt{x^2 - 1}\right] \\ &= 2\tan^{-1}\frac{13}{39} + \tan^{-1}\frac{1}{7} = 2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} \\ &= \tan^{-1}\left\{\frac{2 \times 1/3}{1 - (1/3)^2}\right\} + \tan^{-1}\frac{1}{7} \quad \left[\because 2\tan^{-1}x = \tan^{-1}\frac{2x}{1-x^2}, \text{ if } |x| < 1\right] \\ &= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{7} = \tan^{-1}\left\{\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \times \frac{1}{7}}\right\} = \tan^{-1}1 = \frac{\pi}{4} \end{aligned}$$

5. EQUATION AND IDENTITIES INVOLVING ITF:

Illustration 39

Solve $\sin^{-1}(x^2 - 2x + 1) + \cos^{-1}(x^2 - x) = \frac{\pi}{2}$

Solution:

$$\sin^{-1}(f(x)) + \cos^{-1}(g(x)) = \frac{\pi}{2} \Leftrightarrow f(x) = g(x) \text{ and } -1 \leq f(x), g(x) \leq 1$$

$$x^2 - 2x + 1 = x^2 - x \Leftrightarrow x = 1, \text{ accepted as a solution}$$

Illustration 40

Find the value of $\cos(2\cos^{-1}x + \sin^{-1}x)$ when $x = \frac{1}{5}$

Solution:

$$\begin{aligned} \cos\left(2\cos^{-1}\frac{1}{5} + \sin^{-1}\frac{1}{5}\right) &= \cos\left(\cos^{-1}\frac{1}{5} + \sin^{-1}\frac{1}{5} + \cos^{-1}\frac{1}{5}\right) \\ &= \cos\left(\frac{\pi}{2} + \cos^{-1}\frac{1}{5}\right) = -\sin\left(\cos^{-1}\left(\frac{1}{5}\right)\right) \quad \dots(i) \\ &= -\sqrt{1 - \left(\frac{1}{5}\right)^2} = -\frac{2\sqrt{6}}{5}. \end{aligned}$$

Aliter: Let $\cos^{-1}\frac{1}{5} = \theta \Rightarrow \cos\theta = \frac{1}{5}$ and $\theta \in \left(0, \frac{\pi}{2}\right)$

$$\therefore \sin\theta = \frac{\sqrt{24}}{5}$$

$$\therefore \sin^{-1}(\sin\theta) = \sin^{-1}\left(\frac{\sqrt{24}}{5}\right) \quad \dots(ii)$$

Inverse Trigonometric Functions

$$\therefore \theta \in \left(0, \frac{\pi}{2}\right)$$

$$\Rightarrow \sin^{-1}(\sin \theta) = \theta$$

$\therefore$ equation (ii) can be written as

$$\theta = \sin^{-1} \left(\frac{\sqrt{24}}{5} \right)$$

$$\therefore \theta = \cos^{-1} \left(\frac{1}{5} \right)$$

$$\Rightarrow \cos^{-1} \left(\frac{1}{5} \right) = \sin^{-1} \left(\frac{\sqrt{24}}{5} \right)$$

Now equation (i) can be written as $y = -\sin \left\{ \sin^{-1} \left(\frac{\sqrt{24}}{5} \right) \right\}$... (iii)

$$\therefore \frac{\sqrt{24}}{5} \in [-1, 1] \quad \therefore \sin \left\{ \sin^{-1} \left(\frac{\sqrt{24}}{5} \right) \right\} = \frac{\sqrt{24}}{5}$$

$\therefore$ from equation (iii), we get $y = -\frac{\sqrt{24}}{5}$

Illustration 41

Find value of x if $\cos^{-1}(-x) + \tan^{-1}(-x) - 2 \sin^{-1}(x) + \sec^{-1}\left(-\frac{1}{x}\right) = \frac{\pi}{4}$ for $|x| \leq 1$.

Solution:

$$\pi - \cos^{-1}(x) - \tan^{-1}(x) - 2 \sin^{-1}(x) + \cos^{-1}(-x) = \frac{\pi}{4}$$

$$\pi - \cos^{-1}(x) - \tan^{-1}(x) - 2 \sin^{-1}(x) + \pi - \cos^{-1}(x) = \frac{\pi}{4}$$

$$2\pi - 2(\sin^{-1}x + \cos^{-1}x) - \frac{\pi}{4} = \tan^{-1}x$$

$$2\pi - \pi - \frac{\pi}{4} = \tan^{-1}x \Rightarrow \tan^{-1}x = \frac{3\pi}{4}. \text{ Hence no solution}$$

Illustration 42

The value of 'a' for which $ax + \sin^{-1}(4x - x^2 - 5) + \cos^{-1}(4x - x^2 - 5) = 0$ has a real solution, is -

(A) $-\frac{\pi}{2}$

(B) $-\frac{\pi}{4}$

(C) $\frac{\pi}{4}$

(D) infinitely many solutions

Ans. (B)

Solution:

Given is

$$ax + \sin^{-1}\{-1 - (x-2)^2\} + \cos^{-1}\{-1 - (x-2)^2\} = 0$$

$$\therefore -1 \leq -1 - (x-2)^2 \leq 1 \Rightarrow x = 2$$

$$\text{so, } 2a + \frac{\pi}{2} = 0 \Rightarrow a = -\frac{\pi}{4}$$

Illustration 43

The value of x that satisfies $\tan^{-1}(\tan 3) = \tan^2 x$ is -

- (A) $3 - \pi$ (B) $\sqrt{\pi - 3}$ (C) $\sqrt{2\pi - 3}$ (D) not possible

Ans. (D)

Solution:

$$\tan^2 x = \tan^{-1}(\tan 3)$$

$$\tan^2 x = \tan^{-1}(\tan(3 - \pi))$$

$$\tan^2 x = 3 - \pi$$

but $3 - \pi$ is (-ve)

hence not possible.

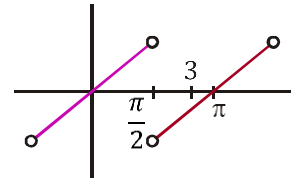


Illustration 44

Number of values of x satisfying $\tan^{-1} x + \tan^{-1}(1-x) = \frac{3\pi}{16} \tan\left(2 \tan^{-1} \frac{1}{2}\right)$ is-

- (A) 0 (B) 1 (C) 2 (D) Infinite

Ans.(C)

Solution:

$$\Rightarrow \tan^{-1} x + \tan^{-1}(1-x) = \frac{3\pi}{16} \left(\frac{2 \cdot \frac{1}{2}}{1 - \frac{1}{4}} \right) = \frac{\pi}{4}$$

$$\Rightarrow \frac{x+1-x}{1-x+x^2} = 1 \Rightarrow x = 0, 1$$

Illustration 45

If $\tan^{-1} y = 4 \tan^{-1} x$, $\left(|x| < \tan \frac{\pi}{8}\right)$, find y as an algebraic function of x and hence prove that $\tan \frac{\pi}{8}$ is a root

of the equation $x^4 - 6x^2 + 1 = 0$.

Solution:

We have $\tan^{-1} y = 4 \tan^{-1} x$

$$\Rightarrow \tan^{-1} y = 2 \tan^{-1} \frac{2x}{1-x^2} \quad (\text{as } |x| < 1)$$

$$= \tan^{-1} \frac{\frac{4x}{1-x^2}}{1 - \frac{4x^2}{(1-x^2)^2}} = \tan^{-1} \frac{4x(1-x^2)}{x^4 - 6x^2 + 1} \quad \left(\text{as } \left| \frac{2x}{1-x^2} \right| < 1 \right)$$

$$\Rightarrow y = \frac{4x(1-x^2)}{x^4 - 6x^2 + 1}$$

$$\text{If } x = \tan \frac{\pi}{8}$$

$$\Rightarrow \tan^{-1} y = 4 \tan^{-1} x = \frac{\pi}{2} \Rightarrow y \text{ is not defined} \Rightarrow x^4 - 6x^2 + 1 = 0$$

Inverse Trigonometric Functions

Illustration 46

Let ${}^{50}C_{15} + 5 {}^{50}C_{14} + 10 {}^{50}C_{13} + 10 {}^{50}C_{12} + 5 {}^{50}C_{11} + {}^{50}C_{10} = {}^{55}C_p$

and ${}^5C_5 + {}^6C_5 + {}^7C_5 + \dots + {}^{54}C_5 = {}^{55}C_q, p, q < 30$.

On the basis of above information, answer the following questions:

- (i) $\sin^{-1} \sin\left((p+q)\frac{\pi}{4}\right) + \cos^{-1} \cos(p) + \tan^{-1} \tan(q)$ equals
- (A) $\frac{13\pi}{4} - 9$ (B) $21 - \frac{25\pi}{4}$ (C) $\frac{11\pi}{4} - 9$ (D) $\frac{27\pi}{4} - 21$
- (ii) If $\sin^{-1}(x - x^2 + x^3 - x^4 + \dots \infty) + \cos^{-1}(x^2 + x^3 + x^4 + \dots \infty) = (p - 2q - 1)\frac{\pi}{4}, (|x| < 1)$ then number of possible real value(s) of x is
- (A) 1 (B) 2 (C) 3 (D) more than 3
- (iii) If $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{q\pi}{2(p-3)}$, then sum of all possible real value(s) of x is
- (A) -1 (B) $-\frac{5}{6}$ (C) 0 (D) $\frac{1}{6}$

Solution:

$${}^5C_0 {}^{50}C_{15} + {}^5C_1 {}^{50}C_{14} + {}^5C_2 {}^{50}C_{13} + {}^5C_3 {}^{50}C_{12} + {}^5C_4 {}^{50}C_{11} + {}^5C_5 {}^{50}C_{10} = {}^{55}C_{15}$$

$$\therefore p = 15$$

$${}^5C_5 + {}^6C_5 + {}^7C_5 + \dots + {}^{54}C_5 = {}^{55}C_6$$

$$\therefore q = 6$$

(i) **Ans.(B)**

$$\sin^{-1} \sin(15+6)\frac{\pi}{4} = \sin^{-1} \sin \frac{21\pi}{4} = \sin^{-1} \sin\left(5\pi + \frac{\pi}{4}\right) = -\frac{\pi}{4}$$

$$\cos^{-1} \cos(15) = 15 - 4\pi$$

$$\tan^{-1} \tan(6) = 6 - 2\pi$$

(ii) **Ans.(A)**

$$\sin^{-1}(x - x^2 + x^3 - x^4 + \dots \infty) + \cos^{-1}(x^2 + x^3 + x^4 + \dots \infty) = \frac{\pi}{2}$$

$$\sin^{-1}\left(\frac{x}{1+x}\right) + \cos^{-1}\left(\frac{x^2}{1-x}\right) = \frac{\pi}{2} \Rightarrow \frac{x}{1+x} = \frac{x^2}{1-x}$$

$$x = 0 \text{ or } \frac{1}{1+x} = \frac{x}{1-x}$$

$$\Rightarrow 1-x = x^2 + x$$

$$\Rightarrow x^2 + 2x - 1 = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{8}}{2}$$

$$\Rightarrow x = -1 \pm \sqrt{2}$$

$$\Rightarrow x = -1 + \sqrt{2}, x = -1 - \sqrt{2} \quad (\text{rejected})$$

so, one value of x .

(iii) Ans.(D)

$$\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$$

$$\Rightarrow \frac{2x+3x}{1-6x^2} = 1$$

$$\Rightarrow 6x^2 + 5x - 1 = 0$$

$$\Rightarrow x = \frac{1}{6}, x = -1 \text{ (rejected)}$$

so, one value of x .

6. SIMULTANEOUS INEQUATIONS AND INEQUATIONS INVOLVING I.T.F.:

For $|x| \leq 1, |y| \leq 1$

$$x > y \Leftrightarrow \begin{cases} \sin^{-1} x > \sin^{-1} y \\ \cos^{-1} x < \cos^{-1} y \end{cases}$$

and for $x, y \in R$

$$x > y \Leftrightarrow \tan^{-1} x > \tan^{-1} y$$

Illustration 47

The solution of the inequality $\pi - \sin^{-1} \sin \left(\frac{2x^2+3}{x^2+1} \right) \leq 3$ -

(A) $[2, 3]$

(B) $[2, 3]$

(C) R^+

(D) R

Ans. (D)

Solution:

$$\frac{2x^2+3}{x^2+1} = \frac{2(x^2+1)+1}{x^2+1} = 2 + \frac{1}{x^2+1}$$

$$0 < \frac{1}{x^2+1} \leq 1 \Rightarrow 2 < 2 + \frac{1}{x^2+1} \leq 3$$

$$\sin^{-1} \sin \left(\frac{2x^2+3}{x^2+1} \right) = \pi - \frac{2x^2+3}{x^2+1}$$

$$\text{Given inequality becomes, } \pi - \left(\pi - \frac{2x^2+3}{x^2+1} \right) \leq 3$$

$$2x^2 + 3 \leq 3x^2 + 3$$

$$x^2 \geq 0 \Rightarrow x \in R.$$

Illustration 48

Find the complete solution set of $\sin^{-1}(\sin 5) > x^2 - 4x$.

Solution:

$$\sin^{-1}(\sin 5) > x^2 - 4x \Rightarrow \sin^{-1}[\sin(5 - 2\pi)] > x^2 - 4x$$

$$\Rightarrow x^2 - 4x < 5 - 2\pi \Rightarrow x^2 - 4x + (2\pi - 5) < 0$$

$$\Rightarrow 2 - \sqrt{9-2\pi} < x < 2 + \sqrt{9-2\pi} \Rightarrow x \in (2 - \sqrt{9-2\pi}, 2 + \sqrt{9-2\pi})$$

Inverse Trigonometric Functions

Illustration 49

Find the complete solution set of $[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$, where $[\cdot]$ denotes the greatest integer function.

Solution:

$$[\cot^{-1}x]^2 - 6[\cot^{-1}x] + 9 \leq 0$$

$$\Rightarrow ([\cot^{-1}x] - 3)^2 \leq 0 \Rightarrow [\cot^{-1}x] = 3 \Rightarrow 3 \leq \cot^{-1}x < 4 \Rightarrow x \in (-\infty, \cot 3]$$

Illustration 50

If $\cot^{-1} \frac{n}{\pi} > \frac{\pi}{6}$, $n \in N$, then the maximum value of n is -

(A) 1

(B) 5

(C) 9

(D) none of these

Ans. (B)

Solution:

$$\cot^{-1} \frac{n}{\pi} > \frac{\pi}{6}$$

$$\Rightarrow \cot \left(\cot^{-1} \left(\frac{n}{\pi} \right) \right) < \cot \left(\frac{\pi}{6} \right) \Rightarrow \frac{n}{\pi} < \sqrt{3}$$

$$\Rightarrow n < \pi\sqrt{3} \Rightarrow n < 5.5 \text{ (approx.)}$$

$$\Rightarrow n = 5 \quad \because (n \in N)$$

7. Summation of Series Involving $\tan^{-1} \left(\frac{x-y}{1+xy} \right) = \tan^{-1}x - \tan^{-1}y$:

Illustration 51

If $A = 2\tan^{-1}(2\sqrt{2} - 1)$ and $B = 3\sin^{-1}(1/3) + \sin^{-1}(3/5)$, then show $A > B$.

Solution:

$$\text{We have, } A = 2\tan^{-1}(2\sqrt{2} - 1) = 2\tan^{-1}(1.828)$$

$$\Rightarrow A > 2\tan^{-1}(\sqrt{3}) \Rightarrow A > \frac{2\pi}{3} \quad \dots(i)$$

$$\text{also, we have, } \sin^{-1} \left(\frac{1}{3} \right) < \sin^{-1} \left(\frac{1}{2} \right) \Rightarrow \sin^{-1} \left(\frac{1}{3} \right) < \frac{\pi}{6}$$

$$\Rightarrow 3\sin^{-1} \left(\frac{1}{3} \right) < \frac{\pi}{2}$$

$$\text{also, } 3\sin^{-1} \left(\frac{1}{3} \right) = \sin^{-1} \left(3 \cdot \frac{1}{3} - 4 \left(\frac{1}{3} \right)^3 \right) = \sin^{-1} \left(\frac{23}{27} \right) = \sin^{-1}(0.852)$$

$$\Rightarrow 3\sin^{-1}(1/3) < \sin^{-1}(\sqrt{3}/2) \Rightarrow 3\sin^{-1}(1/3) < \pi/3$$

$$\text{also, } \sin^{-1}(3/5) = \sin^{-1}(0.6) < \sin^{-1}(\sqrt{3}/2) \Rightarrow \sin^{-1}(3/5) < \pi/3$$

$$\text{Hence, } B = 3\sin^{-1}(1/3) + \sin^{-1}(3/5) < \frac{2\pi}{3} \quad \dots(ii)$$

From (i) and (ii), we have $A > B$.

Illustration 52

The value of $\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{1}{n^2 - n + 1} \right)$ is $\frac{\pi}{a}$, then 'a' is equal to

Ans. (2)

Solution:

$$T_n = \tan^{-1} \left(\frac{1}{1 + n(n-1)} \right) = \tan^{-1} \left(\frac{n - (n-1)}{1 + n(n-1)} \right)$$

$$T_n = \tan^{-1}(n) - \tan^{-1}(n-1)$$

$$T_1 = \tan^{-1}(1) - \tan^{-1}(0)$$

$$T_2 = \tan^{-1}(2) - \tan^{-1}(1)$$

$$T_n = \tan^{-1}(n) - \tan^{-1}(n-1)$$

$$S_n = \tan^{-1}(n)$$

$$S_{\infty} = \tan^{-1}(\infty) = \frac{\pi}{2}$$

Illustration 53

Prove that: $\tan^{-1} \left(\frac{c_1 x - y}{c_1 y + x} \right) + \tan^{-1} \left(\frac{c_2 - c_1}{1 + c_2 c_1} \right) + \tan^{-1} \left(\frac{c_3 - c_2}{1 + c_3 c_2} \right) + \dots + \tan^{-1} \left(\frac{c_n - c_{n-1}}{1 + c_n c_{n-1}} \right) + \tan^{-1} \left(\frac{1}{c_n} \right) = \tan^{-1} \left(\frac{x}{y} \right)$

Solution:

L.H.S.

$$\begin{aligned} & \tan^{-1} \left(\frac{c_1 x - y}{c_1 y + x} \right) + \tan^{-1} \left(\frac{c_2 - c_1}{1 + c_2 c_1} \right) + \tan^{-1} \left(\frac{c_3 - c_2}{1 + c_3 c_2} \right) + \dots + \tan^{-1} \left(\frac{c_n - c_{n-1}}{1 + c_n c_{n-1}} \right) + \tan^{-1} \left(\frac{1}{c_n} \right) \\ &= \tan^{-1} \left(\frac{\frac{x}{y} - \frac{1}{c_1}}{1 + \frac{x}{y} \cdot \frac{1}{c_1}} \right) + (\tan^{-1} c_2 - \tan^{-1} c_1) + (\tan^{-1} c_3 - \tan^{-1} c_2) + \dots + (\tan^{-1} c_n - \tan^{-1} c_{n-1}) + \tan^{-1} \left(\frac{1}{c_n} \right) \\ &= \tan^{-1} \left(\frac{x}{y} \right) - \tan^{-1} \left(\frac{1}{c_1} \right) - \tan^{-1} c_1 + \tan^{-1} c_n + \tan^{-1} \left(\frac{1}{c_n} \right) \\ &= \tan^{-1} \left(\frac{x}{y} \right) - (\cot^{-1} c_1 + \tan^{-1} c_1) + (\tan^{-1} c_n + \cot^{-1} c_n) \\ &= \tan^{-1} \left(\frac{x}{y} \right) - \frac{\pi}{2} + \frac{\pi}{2} = \tan^{-1} \left(\frac{x}{y} \right) = \text{R.H.S.} \end{aligned}$$

Illustration 54

If $\sum_{n=1}^x \tan^{-1} \left(\frac{4}{n^2 + 3} \right) = \pi - \tan^{-1} \frac{26}{7}$ is satisfied by $x \in N$, then the value of $\left[\sin^{-1}(\cos x) \right]$ (where $[.]$ denotes greatest integer function), is -

(A) -2

(B) -1

(C) 0

(D) 1

Ans. (B)

Solution:

$$\begin{aligned} \pi - \tan^{-1} \frac{26}{7} &= \sum_{n=1}^x \tan^{-1} \left(\frac{n+1}{2} \right) - \tan^{-1} \left(\frac{n-1}{2} \right) \\ &= \tan^{-1}(1) - \tan^{-1}(0) + \tan^{-1} \left(\frac{3}{2} \right) - \tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1}(2) - \tan^{-1}(1) + \dots + \tan^{-1} \left(\frac{x+1}{2} \right) - \tan^{-1} \left(\frac{x-1}{2} \right) \\ &= \tan^{-1} \left(\frac{x}{2} \right) - \tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{x+1}{2} \right) = \pi + \tan^{-1} \left\{ \frac{\frac{x+1}{2} + \frac{x}{2}}{1 - \frac{(x+1)x}{4}} \right\} - \tan^{-1} \left(\frac{1}{2} \right) \\ &= \pi + \tan^{-1} \left(\frac{4x+2}{4-x(x+1)} \right) - \tan^{-1} \frac{1}{2} \Rightarrow \frac{x^2+9x}{2(x^2-x-5)} = \frac{26}{7} \Rightarrow x = 4 \end{aligned}$$

$$\text{again, } \left[\sin^{-1}(\cos 4) \right] = \left[\frac{\pi}{2} - \cos^{-1}(\cos 4) \right] = \left[\frac{\pi}{2} - 2\pi + 4 \right] = -1$$

Illustration 55

Sum of series $\sum_{n=0}^{\infty} \cot^{-1}(2^{n+1} + 2^{-n})$ is equal to-

- (A) $\frac{\pi}{8}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2} + \tan^{-1} 2$ (D) $\frac{\pi}{4} + \cot^{-1} 2$

Ans. (B)

Solution:

$$\begin{aligned} \sum_{n=0}^{\infty} \cot^{-1}(2^{n+1} + 2^{-n}) &= \sum_{n=0}^{\infty} \tan^{-1} \left(\frac{1}{2^{n+1} + 2^{-n}} \right) = \sum_{n=0}^{\infty} \tan^{-1} \left(\frac{2^{n+1} - 2^n}{1 + 2^n \cdot 2^{n+1}} \right) = \sum_{n=0}^{\infty} \tan^{-1}(2^{n+1}) - \tan^{-1}(2^n) \\ &= (\tan^{-1} 2 - \tan^{-1} 1) + (\tan^{-1} 4 - \tan^{-1} 2) + \dots \dots \dots \infty \\ &= \frac{\pi}{2} - \tan^{-1}(1) = \frac{\pi}{4} \end{aligned}$$

Illustration 56

If $\tan^{-1} \frac{2}{1^2} + \tan^{-1} \frac{2}{2^2} + \tan^{-1} \frac{2}{3^2} + \tan^{-1} \frac{2}{4^2} + \dots \dots \dots \tan^{-1} \frac{2}{17^2} = \pi - \tan^{-1} \frac{p}{q}$, where p & q are co-prime, then

$(p-q)$ is

Ans. (7)

Solution:

$$\begin{aligned} T_n &= \tan^{-1} \frac{2}{n^2} = \tan^{-1} \left[\frac{(n+1) - (n-1)}{1 + (n-1)(n+1)} \right] \\ \sum_{n=1}^n \tan^{-1} \frac{2}{n^2} &= \tan^{-1} 18 + \tan^{-1} 17 - \tan^{-1} 1 = \pi - \tan^{-1} \frac{34}{27} \\ p - q &= 7 \end{aligned}$$

Illustration 57

If $\sum_{r=1}^{\infty} \cot^{-1} \left(\frac{(r+1)^2}{2} \right) = \tan^{-1} x$, then x is equal to -

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) 3

(D) $\frac{3}{2}$

Ans. (C)

Solution:

$$\sum_{r=1}^n \tan^{-1} \left(\frac{2}{(r+1)^2} \right) = \tan^{-1} \left(\frac{(r+2)-r}{1+r(r+2)} \right)$$

$$= (\tan^{-1} 3 - \tan^{-1} 1) + (\tan^{-1} 4 - \tan^{-1} 2) + (\tan^{-1} 5 - \tan^{-1} 3) + ..$$

$$.. + (\tan^{-1}(n+1) - \tan^{-1}(n-1)) + (\tan^{-1}(n+2) - \tan^{-1}n)$$

$$= \tan^{-1}(n+1) + \tan^{-1}(n+2) - \tan^{-1}1 - \tan^{-1}2$$

$$\sum_{r=1}^{\infty} \tan^{-1} \left(\frac{2}{(r+1)^2} \right) = \frac{\pi}{2} + \frac{\pi}{2} - \tan^{-1}1 - \tan^{-1}2 = \tan^{-1}3 = \tan^{-1}x$$

$$\Rightarrow x = 3$$